

Inverse reinforcement learning the reward landscape of DVM in Atlantic Cod

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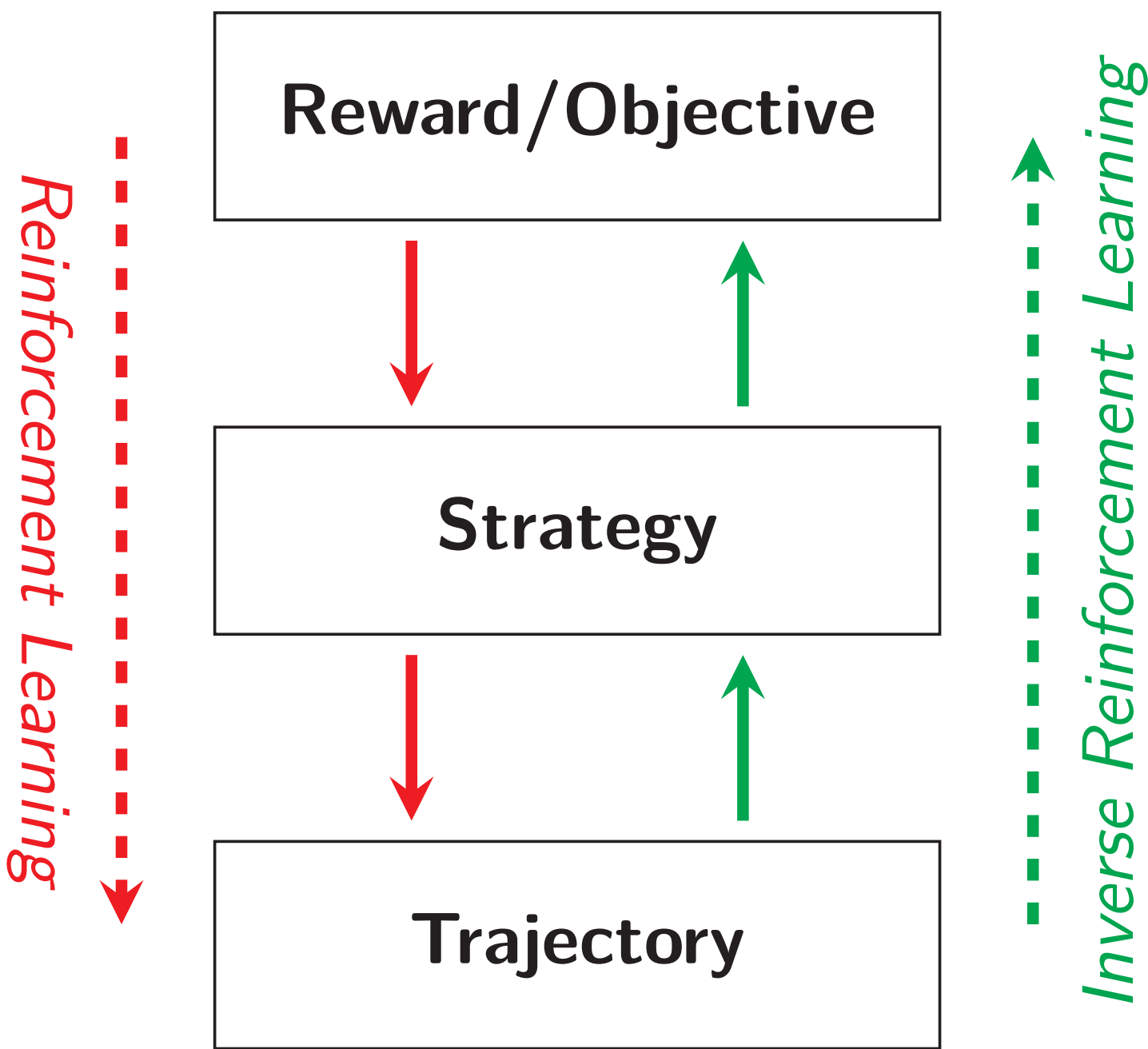
Abstract

Atlantic cod (*Gadus morhua*, L.) migrate up and down the water column every day (**diel vertical migration**, DVM), usually explained as a trade-off between feeding, predation risk and energy conservation. Because these drivers are well studied, DVM is a strong test case for new methods: we know what a sensible result should look like. We used **inverse reinforcement learning** (IRL) to infer from depth tracks alone how rewarding each depth is at each time of day, and can compare the result to established expectations. The behaviour repeats daily, so tracks split naturally into one-day episodes, and depth×time is a comparatively simple state space.

What is Inverse RL?

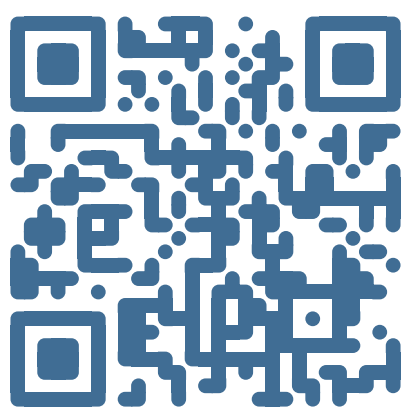
Reinforcement Learning. Given a particular reward, what is the optimal strategy to maximize cumulative reward?

Inverse RL. Given this trajectory, assuming that it is an example of optimal behavior, what reward is it optimizing?



Because the strategy is relative to the environment, combining Inverse RL and "regular" RL is promising: the empirically recovered reward can be used to learn an optimal strategy, the behavioral results of which can be investigated in simulated environments. This is promising to, e.g., investigate impacts of climate change on fish behavior.

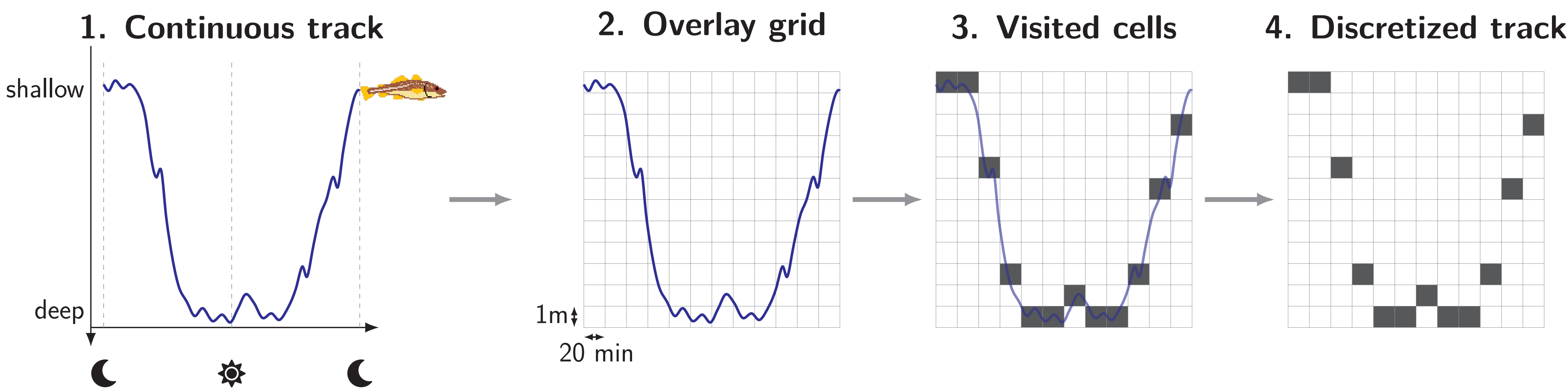
Find more information on the mathematical foundations of Maximum Entropy Inverse Reinforcement Learning and the inference procedure here:



Data Processing & Inference

Dataset. Atlantic Cod from Tvedestrand Fjord (SE Norway), tracked between **2014** and **2020**. **Only depth** positions were considered to enhance data availability and keep the state space low-dimensional, only depths up to 30 m and only complete days of data (midnight to midnight) were used.

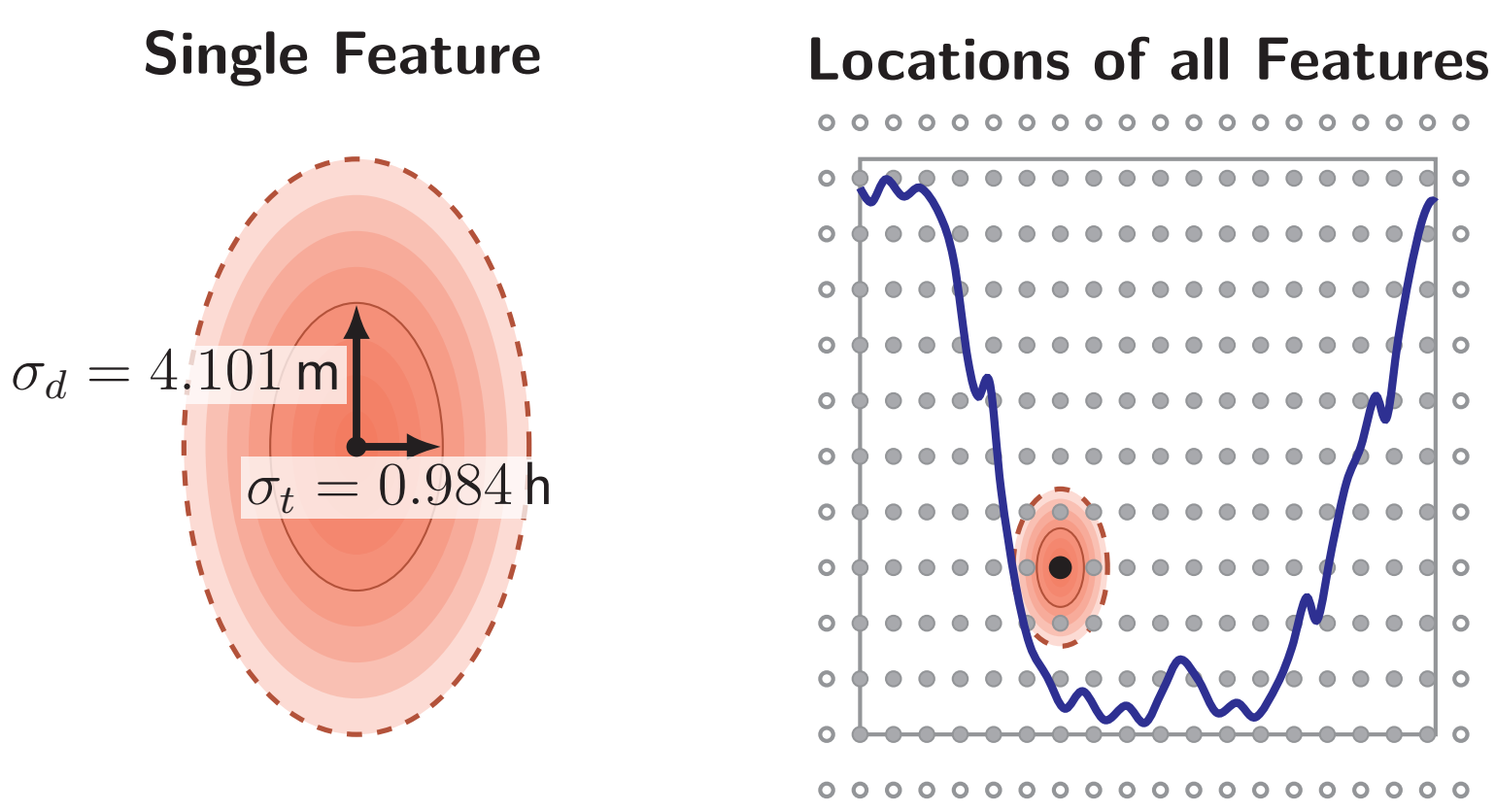
Preprocessing. Individual depth records were **discretized** into grid cells with **20 min temporal resolution** and **1 m vertical resolution** to form **tracks** of one individual-day of depth selections in 72 time steps and 30 depth cells - from **midnight to midnight**.



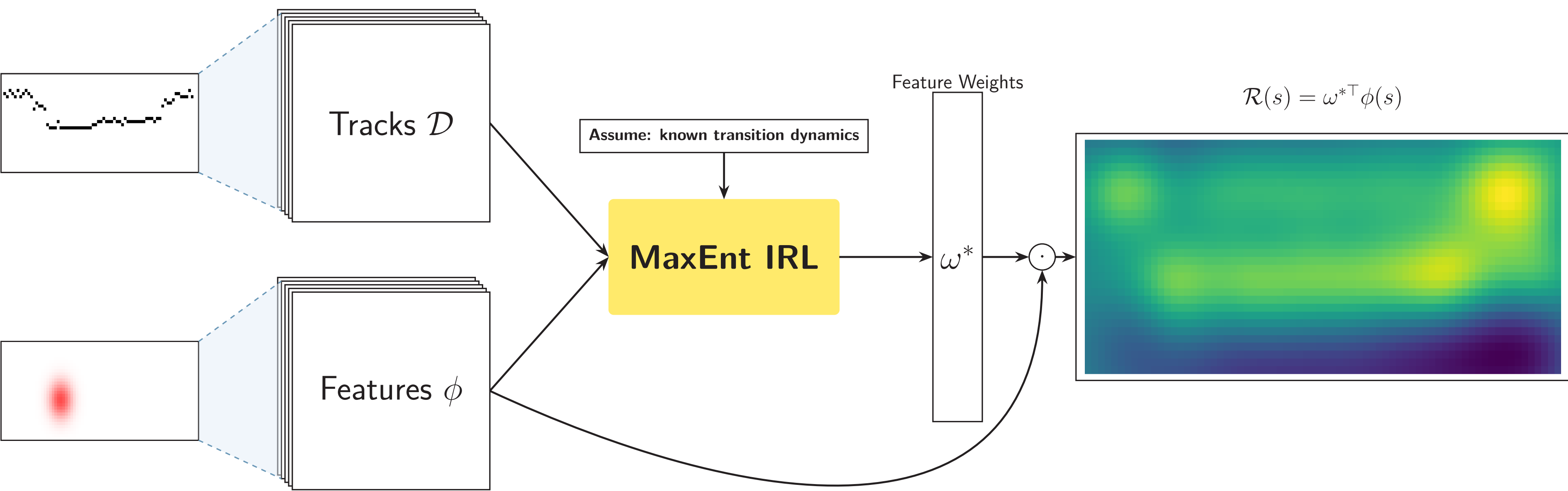
Filtering. The discretized tracks were filtered to match the necessary assumptions of MaxEnt IRL - to keep transitions between states simple, a **maximum depth change of ±5 m** between time steps is required. To demonstrate the ability to recover known behavioral strategies, the tracks were further filtered to **only those where DVM is exhibited** by requiring a minimum distance of 10 m between the shallowest and deepest localisations per day.

Features.

- form basis of reward through learned weights
- evenly spaced radial basis functions
- sum of all features roughly constant across state space
- padding - some features have center outside of state space to keep sum constant towards the edges
- learned weights mirror attractiveness of location



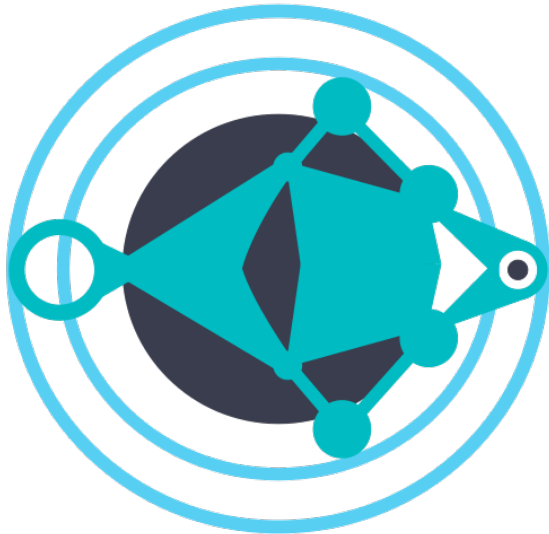
Inference Procedure:



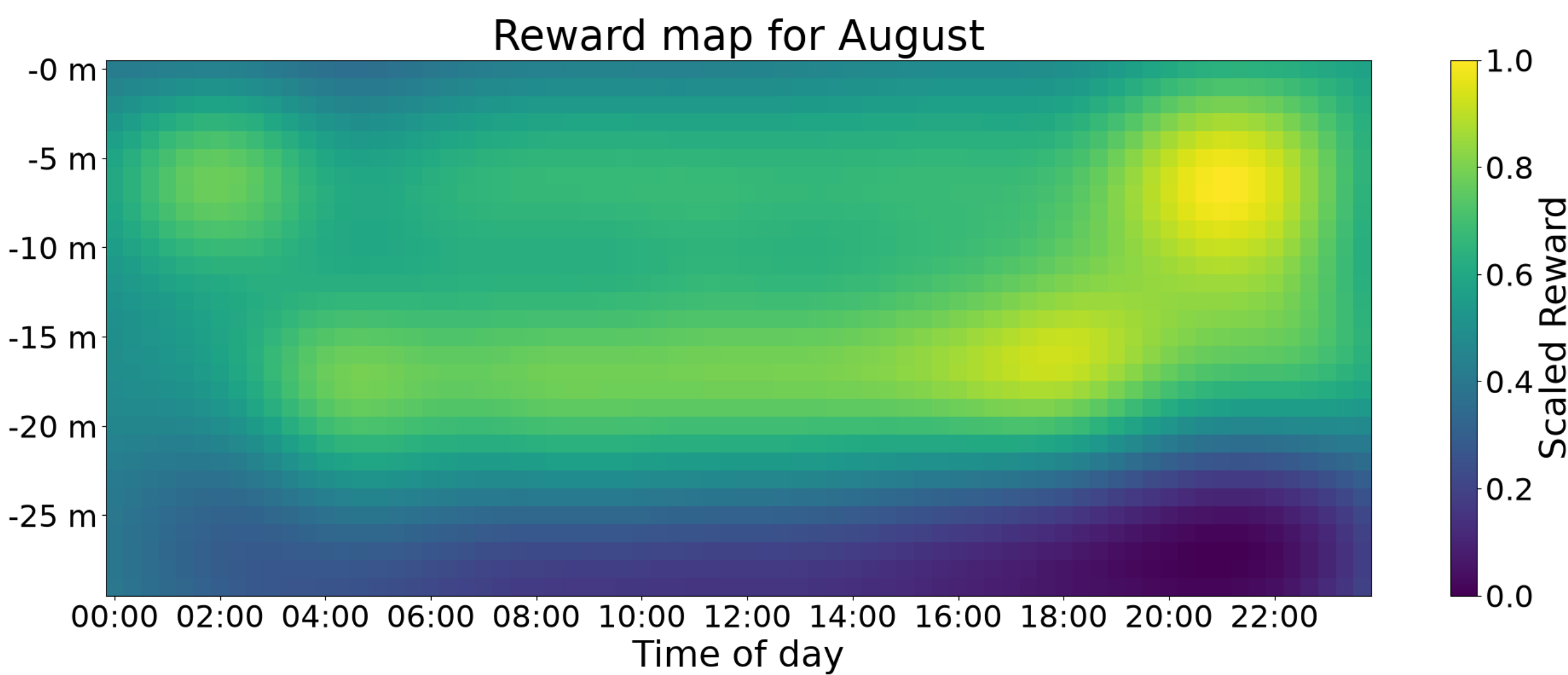
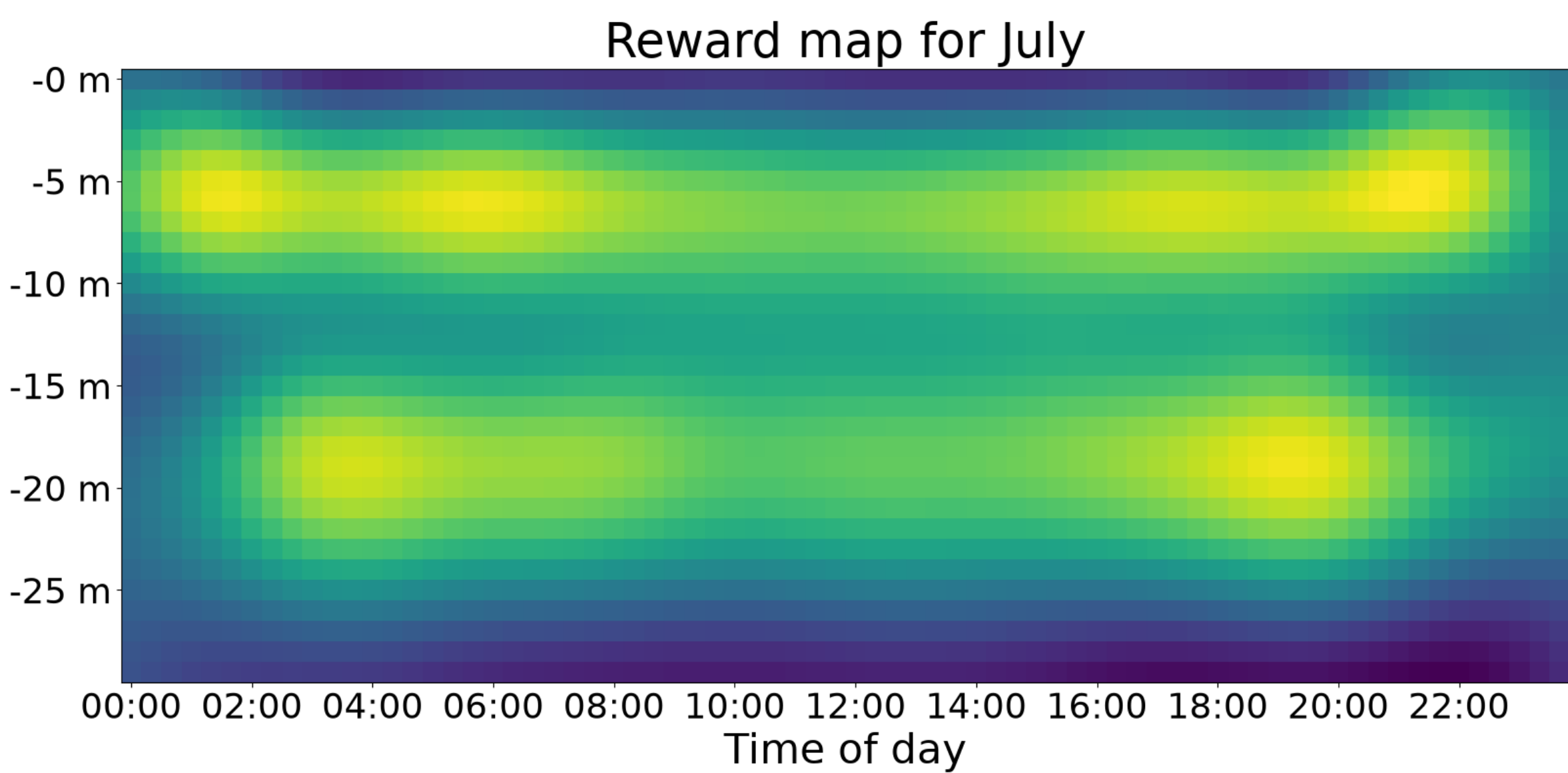
Contact



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Results



What's next? With the learned reward, an optimal strategy (policy) for each month can be calculated, allowing for two things: 1) benchmarking of real tracks relative to the theoretical optimum and 2) simulation of tracks. How do these compare to actual tracks? With some stochasticity, does the recovered reward lead to plausible daily trajectories? How does the policy react to counterfactual rewards, like adding a high reward for shallow depths at noon?